

Rodrigue GOVAN, Ph.D.

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Education

- October 2021 – August 2025 **Ph.D. in Computer Science – Data Science**, University of New Caledonia
- Thesis title: “**Deep Learning on Attributed Graphs for Mapping Leptospirosis Risk**”.
 - Supervision: Prof. Nazha SELMAOUI-FOLCHER (University of New Caledonia) and Prof. Philippe FOURNIER-VIGER (Shenzhen University, China).
 - Fundings: Doctoral scholarship (50%), ANR-SPIraL program (ANR-19-CE35-0006-02) by the French National Research Agency (50%).
 - Jury members:
 - Christophe MENKÈS, *Senior Research Scientist* Chair
ENTROPIE, French National Institute of Research for Sustainable Development (IRD)
 - Thomas GUYET, *Full Researcher* Reviewer
AISToSight, Inria Research Centre, Lyon, France
 - Luiz-Angelo STEFFENEL, *Full Professor* Reviewer
LICIIS, University of Reims Champagne-Ardenne
 - Cyrille GOARANT, *Research Scientist* Examiner
Public Health Department, South Pacific Community
 - Corina IOVAN, *Research Scientist* Examiner
ENTROPIE, French National Institute of Research for Sustainable Development (IRD)
 - Nadia KABACHI, *Full Professor* Examiner
ERIC, University Claude Bernard Lyon 1, France
 - Nazha SELMAOUI-FOLCHER, *Full Professor* Supervisor
ISEA, University of New Caledonia
 - Philippe FOURNIER-VIGER, *Distinguished Professor* Supervisor
Big Data Institute, CSSE, Shenzhen University, China
- 2017 – 2019 **M.Sc. in Applied Mathematics and Statistics**, University of Bordeaux
- Minor: Statistical and Stochastic Modeling.
 - Final-year project: “**Object detection in satellites images using deep learning**”.
- 2014 – 2017 **B.Sc. in Applied Mathematics and Computer Science**, University of Bordeaux
- Minor: Business Administration.
 - Final-year project: “**Object-oriented programming of a role-playing game in Java**”.

Experiences

- September 2020 – December 2020 **Data Scientist**, Institute of Exact and Applied Sciences, UNC – Noumea, New Caledonia
- Analysis of natural water quality data in New Caledonia.
 - Contributions to research paper writing.
 - Tools used: Python (`sklearn`, `statsmodels`, `plotly`, `Jupyter`), QGIS, Microsoft Access, PostgreSQL, L^AT_EX.

- August 2019 – February 2019 **Data Scientist Intern**, IDAIA Group – Bruges (Bordeaux), France
 - Deep Learning methods for object detection on satellite images.
 - Over 200M images (covering the entire country of France) processed.
 - Tools used: Python (**sklearn**, **Tensorflow**, **PyTorch**, **plotly**, **tensorboard**, **WandB.ai**, **Jupyter Notebook**), QGIS, Linux, **L^AT_EX**.

Teaching activities

Teachings

Course name	Level	Tutorials	Labs	Total	Years
Algorithms and Programming II in Python	B.Sc. CS 1 st year	×	34	34h	2022
Graph theory and Algorithms	B.Sc. CS 2 nd year	12h	48h	60h	2022, 2023, 2024
Advanced databases	B.Sc. CS 2 nd year	×	48h	48h	2022, 2023, 2024
Introduction to data analysis	B.Sc. CS 3 rd year	4h	×	4h	2023
Applied statistics for geosciences	B.Sc. Geology 1 st year	8h	×	8h	2024
Total		24h	130h	154h	

Internship supervision

- November 2024 – January 2025 **Front-End Developer** – Waren DERAMANE (2nd year B.Sc. in Computer Science)
 - Design and development of a graphical user interface for managing and visualizing spatio-temporal data.
 - Supervision: 80% R. GOVAN, 20% N. SELMAOUI-FOLCHER.
- November 2024 – January 2025 **Back-End Developer** – Saïden TIDJINE (2nd year B.Sc. in Computer Science)
 - Design and development of a spatio-temporal database to support data analysis and management.
 - Supervision: 80% R. GOVAN, 20% N. SELMAOUI-FOLCHER.

Technical skills

Operating systems	macOS, Linux (Ubuntu, Raspberry Pi OS, Debian), Windows
Management and deployment tools	SLURM workload manager, Docker, GitHub/GitLab
Databases	MySQL, PostgreSQL/PostGIS
Programming languages	Python, R, MATLAB/Scilab, Java, HTML, CSS, SQL, PHP, Shell Unix, L^AT_EX
Specific librairies	PyTorch, scikit-learn, dash, rasterio
Proficiency in	Machine learning, Data mining, Databases, Data cleaning and preparation, Implementation and deployment of dashboards

Projects

- 18–20 June 2021 **Hackathon (CASSINI.eu, 1st Edition)** – University of New Caledonia
 - European entrepreneurship-oriented Hackathon.

- Hackathon's theme: "Digitizing Green Spaces".
- Proposed project: SANDLESS (Software to **A**nalyze **N**atural **D**ata to **L**ower **E**cological **S**tress with **S**atellites), a software analyzing satellites, social, economical and environmental data of cities from all over the world in order to help decision-makers to build their own city.
- French rank: 1st.
- European (final) rank: 3rd.

- October 2018 **Big Data Project** – University of Bordeaux
 – January 2019
- Kaggle-like contest organized by the INSA-Toulouse engineering school in collaboration with Airbus Defence and Space.
 - Contest including universities and engineering schools in France.
 - Objective: To determine, by using satellite images, whether or not the image contains a wind turbine.
 - Tools used: Python, Tensorflow, Keras.
 - Final rank: 7/64 teams (with a 97.7% accuracy).

Publications and Communications

Journal papers with peer review

- [3] Govan, R., Scherrer, R., Fougeron, B., Laporte-Magoni, C., Thibeaux, R., Genthon, P., Fournier-Viger, P., Goarant, C., and Selmaoui-Folcher, N. (2025). **Spatio-temporal risk prediction of leptospirosis: A machine-learning-based approach**. *PLOS Neglected Tropical Diseases*, 19(1), e0012755.
DOI: [10.1371/journal.pntd.0012755](https://doi.org/10.1371/journal.pntd.0012755)
- [2] Thibeaux, R., Genthon, P., Govan, R., Selmaoui-Folcher, N., Tramier, C., Kainiu, M., Soupé-Gilbert, M.-E., Wijesuriya, K., and Goarant, C. (2024). **Rainfall-driven resuspension of pathogenic Leptospira in a leptospirosis hotspot**. *Science of The Total Environment*, 911, 168700.
DOI: [10.1016/j.scitotenv.2023.168700](https://doi.org/10.1016/j.scitotenv.2023.168700)
- [1] Scherrer, R., Govan, R., Quiniou, T., Jauffrais, T., Lemonnier, H., Bonnet, S., and Selmaoui-Folcher, N. (2022). **Real-Time Automatic Plankton Detection, Tracking and Classification on Raw Hologram**. In: *International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics* (pp. 25-39). Springer, Cham.
DOI: [10.1007/978-3-031-20837-9_3](https://doi.org/10.1007/978-3-031-20837-9_3)

International conferences with peer review

- [5] Govan, R., Scherrer, R., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **SPAPOOL: Soft Partition Assignment Pooling for Graph Neural Networks**. In: Leung, C.K., Dignös, A., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Big Data Analytics and Knowledge Discovery. DaWaK 2025*. Lecture Notes in Computer Science, vol 16048. Springer, Cham.
DOI: [10.1007/978-3-032-02215-8_27](https://doi.org/10.1007/978-3-032-02215-8_27)
- [4] Govan, R., Selmaoui-Folcher, N., Giannakos, A., and Fournier-Viger, P. (2023). **Co-location Pattern Mining Under the Spatial Structure Constraint**. In: Strauss, C., Amagasa, T., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Database and Expert Systems Applications. DEXA 2023*. Lecture Notes in Computer Science, vol 14146. Springer, Cham.
DOI: [10.1007/978-3-031-39847-6_13](https://doi.org/10.1007/978-3-031-39847-6_13)

- [3] Tokotoko, J., **Govan, R.**, Lemonnier, H., and Selmaoui-Folcher, N. (2022). **Multiscale and Multivariate Time Series Clustering: A New Approach**. In: Ceci, M., Flesca, S., Masciari, E., Manco, G., Raś, Z.W. (eds) *Foundations of Intelligent Systems. ISMIS 2022*. Lecture Notes in Computer Science(), vol 13515. Springer, Cham.
DOI: [10.1007/978-3-031-16564-1_27](https://doi.org/10.1007/978-3-031-16564-1_27)
- [2] Scherrer, R., **Govan, R.**, Quiniou, T., Jauffrais, T., Lemonnier, H., Bonnet, S., and Selmaoui-Folcher, N. (2021). **Automatic Plankton Detection and Classification on Raw Hologram with a Single Deep Learning Architecture**. In: *CIBB 2021 Computational Intelligence Methods for Bioinformatics and Biostatistics*.
URL: [hal-03565469](https://hal.archives-ouvertes.fr/hal-03565469)
- [1] Tokotoko, J., Selmaoui-Folcher, N., **Govan, R.**, and Lemonnier, H. (2021). **TSX-Means: An Optimal K Search Approach for Time Series Clustering**. In: Strauss, C., Kotsis, G., Tjoa, A.M., Khalil, I. (eds) *Database and Expert Systems Applications. DEXA 2021*. Lecture Notes in Computer Science(), vol 12924. Springer, Cham.
DOI: [10.1007/978-3-030-86475-0_23](https://doi.org/10.1007/978-3-030-86475-0_23)

National conferences with peer review

- [3] **Govan, R.**, Scherrer, R., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **Pooling de Graph Neural Networks : une approche dense mais adaptative**. In: *CNIA 2025-Conférence Nationale en Intelligence Artificielle*, PFIA (No. 55-63).
URL: [hal-05197596v1](https://hal.archives-ouvertes.fr/hal-05197596v1)
- [2] **Govan, R.**, Scherrer, R., Goarant, C., Cannet, A., Fournier-Viger, P., and Selmaoui-Folcher, N. (2025). **Cartographie du risque épidémiologique : Le défi des données fortement déséquilibrées**. In: *Revue des Nouvelles Technologies de l'Information, 25èmes Journées Francophones Extraction et Gestion des Connaissances, EGC 2025*, vol. RNTI-E-41. (pp. 159-170).
URL: [hal-04945686](https://hal.archives-ouvertes.fr/hal-04945686)
- [1] **Govan, R.**, Selmaoui-Folcher, N., Giannakos, A., and Fournier-Viger, P. (2023). **Extraction de co-localisations sous contrainte de la structure spatiale**. In: *CNIA 2023-Conférence Nationale en Intelligence Artificielle*, PFIA (No. 53-61).
URL: [hal-04945686](https://hal.archives-ouvertes.fr/hal-04945686)

Posters and workshops

- [2] Genthon, P., Thibeaux, R., Selmaoui-Folcher, N., **Govan, R.**, Kainiu, M., Soupé-Gilbert, M. E., and Goarant, C. (2024). **Leptospira in Rivers of a Leptospirosis Hotspot: Scale Effects**. In: *AGU Fall Meeting Abstracts* (Vol. 2024, No. 1055, pp. H13D-1055).
URL: [2024AGUFMH13D.1055G](https://doi.org/10.1029/2024AGUFMH13D.1055G)
- [1] **Govan, R.** (2022). **Evolving and Dynamic Attributed Graphs: Application to the Risk Mapping of Leptospirosis in New Caledonia**. In: *15^{ème} Édition des Doctoriales*. Pacific Doctoral School (ED469). University of New Caledonia.

Project reports and thesis

- [2] **Govan, R.** (2025). **Deep Learning on Attributed Graphs for Mapping Leptospirosis Risk**. Ph.D. thesis. University of New Caledonia.
URL: [tel-05268357](https://tel.archives-ouvertes.fr/tel-05268357)
- [1] **Govan, R.** (2019). **Deep Learning: Residential pool detection in France**. M.Sc. thesis. University of Bordeaux, France.

- [3] Govan, R., Parmentier, J.-B., and Quiniou T. (2025). **Discover an interactive 3D model of New Caledonia**. In : Nuit de la Science, Fête de la Science 2025. Université de la Nouvelle-Calédonie.
- [2] Govan, R. (2023). **La recherche scientifique au service de la *data***. In : 7^{ème} *meetup consacré à la data*. ISI-NC, OoTech. Nouvelle-Calédonie.
- [1] Govan, R. (2022). **Graphes attribués dynamiques et évolutifs : Application à la Cartographie du Risque de Leptospirose en Nouvelle-Calédonie**. In : 15^{ème} *Édition des Doctoriales*. École Doctorale du Pacifique (ED469). Université de la Nouvelle-Calédonie.

Professional and academic references

Prof. Nazha SELMAOUI-FOLCHER	Full Professor in Computer Science. Institute of Exact and Applied Sciences, University of New Caledonia. Ph.D. supervisor. E-mail: nazha.selmaoui@unc.nc
Prof. Philippe FOURNIER-VIGER	Distinguished Professor in Computer Science. Big Data Institute, College of Computer Science and Software Engineering, Shenzhen University, China. Ph.D. supervisor. E-mail: philfv@szu.edu.cn
Prof. Jérémie BIGOT	Full Professor in Applied Mathematics. Institute of Mathematics in Bordeaux, University of Bordeaux, France. M.Sc. supervisor. E-mail: jeremie.bigot@math.u-bordeaux.fr